Confound Auditing in Medical Classifiers: Evidence from TB, Glioma, and Malaria

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Abstract

A single aggregate performance number gives limited evidence that a medical classifier has learned disease rather than a correlate of disease. We propose an iterative falsification audit for frozen-feature classifiers. A confound-stratified test ablates shortcut and residual directions under random-direction and retrained-head controls, decision rules map each verdict to the next test, and a planted-marker calibration reports the audit's own error rates. On the planted marker, 0 of 20 splits were flagged at no association, and marker prevalence in TB cases of 0.7 or higher was flagged in at least 85% of splits for two of three encoders. In tuberculosis (TB) radiographs, disease extent was not supported as the shortcut. Acquisition source was perfectly decodable (AUC = 1.000) but coincides with the label, a Belarus/RSNA-trained detector scored AUC 0.455 to 0.542 on source-matched data, and nonlinear probes still decoded source after linear erasure (MLP AUC 0.941 to 0.998). In glioma, ablating WHO grade collapsed predictions to one class, so the ablation carries no subgroup information, and a grade-excluded refit ranged from AUC 0.563 to 0.812 across imputation choices. In malaria, views with estimated parasite pixels replaced stayed predictive (patient-grouped AUC 0.952 to 0.977), and uninfected cells carried patient infection status (AUC 0.946 to 0.954). In none of the three domains did the first audit result establish whether the classifier was valid.

1. Introduction

Medical image classifiers can reach high AUROC through features that correlate with the label but carry no diagnostic meaning (Zech et al., 2018; DeGrave et al., 2021; Geirhos et al., 2020). Rejecting one named shortcut leaves other label-aligned variables untested, and supporting it does not show whether disease signal remains.

Gap and purpose. Prior work tests one named shortcut or locates a cue once suspected (Section 2). We found no protocol that states what each outcome of a shortcut test licenses next, reports the audit's own false-flag rate and power, and checks blind spots such as linear-only erasure. The purpose of this study is to propose such a protocol, an *iterative falsification audit*, and to examine how it behaves on three public datasets where the shortcut is not known in advance. TB, glioma and malaria serve as case studies of the loop.

Proposal. The audit has three parts (Section 3). A *stratified test* splits held-out samples by shortcut-label concordance and ablates shortcut and residual directions under three controls. *Decision rules* map each verdict (not supported, supported, restricted view still predictive) to a fixed class of follow-up. A *calibration* on a planted marker gives the false-flag rate and power. The choices that fill the gap are the concordant/discordant split, retrained-head and random-direction controls, nonlinear probes after linear erasure, source-matched evaluation and patient-grouped views.

Contributions. To our knowledge, this is the first study to (1) report the false-flag rate and detection power of a representation-level shortcut audit against a planted marker of controlled strength in medical images, across three frozen encoders (Section 5.1), (2) evaluate a TB detector built from Belarus TB Portals cases and RSNA normals on source-matched data (Section 5.2), (3) show that linear erasure of a label-coincident medical-imaging source confound leaves it decodable by nonlinear probes, and that erasers fit in one encoder do not carry over to another (Section 5.2, Appendix J), and (4) test whether uninfected NIH malaria cells carry patient infection status (Section 5.4). We also show that the grade ablation in TCGA glioma reduces to a constant prediction. Every ingredient has precedent, and the "first" statements apply only to our literature search, which was not systematic (Appendix C).

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2. Related Work

Classifiers exploit site identity (Zech et al., 2018), patient attributes (Jabbour et al., 2020), chest drains (Jiménez-Sánchez et al., 2022), and regions outside the lungs (Maguolo and Nanni, 2020). Error auditing finds hidden low-performing subsets without a prior hypothesis (Oakden-Rayner et al., 2020), and a TB detector trained on one population lost accuracy on another (Sathitratanacheewin and Pongpirul, 2018). Shortcut testing relates attribute encoding to unfairness (Brown et al., 2023), layer-wise probes locate drain and scanner cues (Pedersen et al., 2026), synthetic edits test reliance on off-target findings (Cooke et al., 2025), and site signal in brain-MRI embeddings can be projected out (Rahbar, 2026). Post hoc explanations can fail on unknown artifacts (Adebayo et al., 2022). INLP (Ravfogel et al., 2020) and LEACE (Belrose et al., 2023) erase a concept from linear probes only, ablating a probe direction does not test necessity because a refitted probe can recover (Jiang, 2026), and last-layer retraining recovers core features (Kirichenko et al., 2023).

Gaps. Most of these studies test a named or suspected cue. We found none that calibrates a medical-imaging audit on a planted shortcut, checks linear erasure of a source confound with nonlinear probes or tests eraser transfer across encoders, evaluates a TB detector on source-matched data, or tests patient state in uninfected malaria cells (Hou et al., 2026) (gaps G1 to G8, Appendix C).

3. Audit Procedure

Workflow. The audit names a candidate shortcut s and stratifies held-out samples by concordance with it. It then ablates shortcut and residual directions under fixed-head, retrained-head and random-direction controls, uses the verdict to select a targeted follow-up in each domain, and calibrates the audit itself on a planted marker (Figure 2, Appendix B).

Stratified audit. Let z be a frozen feature vector (an encoder output, or the input variables for tabular glioma data), y the label and s a binary candidate shortcut. Held-out samples are *concordant* when s agrees with y as it does in training and *discordant* otherwise, and a classifier that relies on s scores lower on discordant samples. We fit a logistic classifier with weight vector d_c and a logistic probe for s with weight vector d_s. The *residual direction* d_r is the part of d_c orthogonal to d_s. *Ablating* a unit direction $\hat{d}$ replaces z by $z - (z^\top \hat{d})\hat{d}$.

Controls. Because d_r is built from the classifier, removing it is expected to damage that classifier. Every ablation is therefore reported with the original head, with a head retrained on ablated features, and against 200 covariance-matched random directions, which gives an empirical p-value. For a confound with known labels (TB source) we apply LEACE and up to 200 rounds of INLP, then probe held-out data with linear, MLP and gradient-boosting (GBM) probes.

Decision rules. If the tested shortcut is not supported, we search for other variables aligned with the label (source, patient). If it is supported, we test whether disease signal survives without it (refit). If a restricted view stays predictive, we add views and groupings that separate competing explanations (random crop, grayscale, patient-grouped splits). A verdict never certifies validity. It only selects the next hypothesis to falsify.

Calibration and reporting. We plant a bright corner block with a soft gradient in Shenzhen and Montgomery radiographs with $P(\text{marker} \mid \text{TB}) = \rho$, where ρ is the marker prevalence among TB cases (not a correlation coefficient) and $\rho = 0.5$ is the no-association setting. Over 20 splits per ρ the audit flags reliance when erasing the marker raises AUC on marker-flipped test images by at least 0.02 and above the 95th percentile of random directions. Intervals are 95% bootstrap intervals over test samples (patients in the bystander test) and exclude training variance. Ablation controls use 2 to 10 seeds and glioma refits 20 splits.

Deviations from the protocol. The loop was not pre-registered. Follow-up tests were chosen after earlier results with no multiplicity correction, so p-values and intervals are descriptive. Source-matched evaluation was added because the extent test was not supported, and the TB extent ablation was run in two pipelines that disagree (Appendices K and L).

4. Data and Setup

TB. Belarus TB Portals DICOM cases (Rosenthal et al., 2017) (517 pulmonary-only, group A, and 533 with extrapulmonary disease, group B) against

8,851 RSNA PNG normals (Shih et al., 2019). The candidate shortcut is disease extent (A vs. B). Shenzhen and Montgomery radiographs (Jaeger et al., 2014), where TB and normals share an acquisition pipeline, give source-matched evaluation. Features come from torchxrayvision DenseNet-121 (xrv, 1024-d) (Cohen et al., 2022), ResNet-50, DINOv2-small and a harmonized xrv variant.

Glioma and malaria. Tabular TCGA data (Ceccarelli et al., 2016) for 995 patients (1,122 rows minus 127 without IDH status) use WHO grade (Louis et al., 2021) (the shortcut), MGMT, 1p/19q and TERT, with 257 concordant and 141 discordant test patients. NIH thin-smear cell images (Rajaraman et al., 2018) are encoded by ResNet-50, DenseNet-121 and DINOv2-small on six views (full image, parasite region, RBC view with estimated parasite pixels replaced by the median cell colour, random off-centre crop, grayscale, cell silhouette), plus handcrafted colour and shape features. Evaluation is 5-fold cross-validation grouped by 201 patient identifiers parsed from file names (Appendix D).

5. Results

5.1. Calibration on a planted marker

At $\rho = 0.5$ the audit never flagged reliance (0 of 20 splits for each encoder, so the one-sided 95% upper bound on the false-flag rate is 0.14). For DINOv2 and ResNet-50 the flag rate was 0.85 at $\rho = 0.7$ and 1.00 at $\rho = 0.8$ and 0.9 (Figure 1c). The main TB encoder, xrv, was less sensitive, with flag rates of 0.30, 0.75 and 0.70 at $\rho = 0.7$, 0.8 and 0.9, and 0.00 at $\rho = 1.0$. The recovered direction had cosine 0.64 to 0.72 with the planted one for DINOv2 and ResNet-50 and about 0.25 for xrv (Appendix I), so a negative audit on xrv features is weaker evidence, and xrv power is not monotone in ρ.

5.2. TB: disease extent and acquisition source

Extent. In the first pipeline all nine train/test cells met the WHO target product profile (TPP, World Health Organization, 2014, Appendix Table 6) and the sensitivity gap between groups A and B under a pulmonary-only-trained detector was 0.007 (95% CI −0.032 to 0.045). Both groups are Belarus DICOM, so a detector that only recognizes that format would give the same result. In xrv features, ablating the extent direction left positive-class accuracy unchanged but lowered normal-class accuracy from 0.998 to 0.835 ($p = 0.01$ against random directions), and a retrained head restored 0.998. The uncentered ablation also shifts logits by a constant, so a threshold shift explains this pattern and head recovery does not show the direction is dispensable. Residual-direction ablation collapsed positive-class accuracy (xrv 0.108 and 0.236), and a retrained head recovered it. A second pipeline showed no accuracy change under the extent ablation, and the two are not reconciled (Appendix K).

Source. Source is decodable at AUC 1.000 (DINOv2, ResNet-50) and 0.999 (xrv). Every Belarus image is TB and every RSNA image normal, so source decodability equals TB decodability, and only source-matched data and metadata-only probes can separate format from pathology. The Belarus/RSNA-trained detector scored AUC 0.455 (95% CI 0.417 to 0.499), 0.511 (0.471 to 0.552) and 0.542 (0.503 to 0.584) for DINOv2, ResNet-50 and xrv on Shenzhen and Montgomery combined (Figure 1b). Within-source detection reached 0.893 to 0.903 in Shenzhen. Shenzhen and Montgomery also differ in population from the Belarus and RSNA sources, so format alone cannot be blamed (Appendix F). Harmonizing format raised the Belarus/RSNA detector to 0.752 (0.715 to 0.785) on matched data while source stayed decodable (0.999), so decodability did not track harm. Image metadata alone reached 0.748 within the matched sets, so matching does not remove every label-aligned variable.

Erasure. LEACE reduced the linear source probe to chance in one step (0.470 to 0.494 across feature sets). MLP probes still reached 0.989 to 0.998 and GBM probes 0.985 to 1.000. INLP needed 100 to 200 rounds to reach linear chance, and MLP AUC after 200 rounds was 0.941 to 0.994 (Figure 1a, Appendix F). A linear-only audit would therefore report source as removed while nonlinear decodability remains. Matched-data detection did not change after erasure (xrv 0.808 to 0.825).

5.3. Glioma: grade ablation reduces to a constant

With grade as input the classifier reached concordant AUC 1.000 and discordant AUC 0.000, as the cell

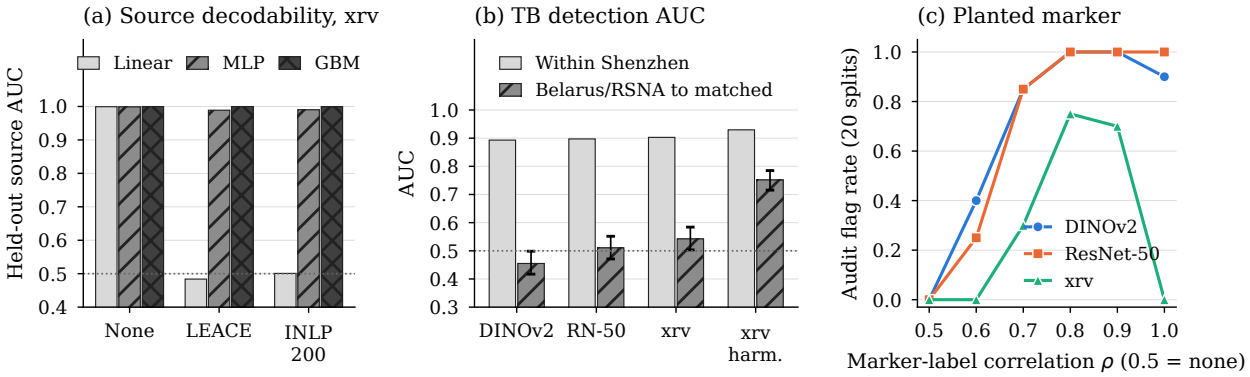


Figure 1: (a) Held-out AUC of source probes after erasure of source in xrv features (chance 0.5). (b) TB detection AUC within Shenzhen (cross-validated) and for the Belarus/RSNA-trained detector on Shenzhen and Montgomery, with 95% bootstrap intervals. (c) Fraction of 20 splits in which the audit flags reliance on a planted marker, by marker prevalence ρ in TB cases.

definition implies, and the grade direction had cosine 1.00 with the classifier direction. Ablating it made the classifier predict IDH-mutant for every patient (accuracy 0.490 on concordant patients, 0.000 on IDH-wildtype low-grade and 1.000 on IDH-mutant glioblastoma patients). Subgroup accuracy therefore reflects the class prior.

The grade-excluded refit carries the glioma conclusion. It reached concordant AUC 0.877 (sd 0.014 over 20 splits) and discordant AUC 0.738 (95% CI 0.649 to 0.823). Median imputation gave 0.812, median imputation with missingness flags 0.563 (0.394 to 0.596), dropping 1p/19q 0.624, and missingness indicators alone 0.666. Co-deletion of 1p/19q implies IDH mutation by definition (Louis et al., 2021), so part of the grade-independent signal is definitional. We support only that modest IDH signal remains without grade and that its size is not stable (Appendix G).

5.4. Malaria: signal after parasite-pixel replacement, including patient state

Patient-grouped AUC was close to cell-level AUC (0.977 vs. 0.982 for ResNet-50), so patient overlap does not explain the high values, although grouping would not remove a slide-level cue shared across patients (Appendix H).

Staining. A saturation-defined staining ablation left ResNet-50 accuracy at 0.927 against 0.928 ($p = 0.95$), and grayscale (0.975 to 0.989) and colour-only features (0.967 to 0.969) stayed near the full image (0.977 to 0.991). Neither isolates a stain effect, so Giemsa staining is not excluded.

Parasite-removed views. The RBC view replaces mask pixels with a flat patch that can itself mark infected cells, so it does not show signal outside the parasite. It reached AUC 0.952 to 0.977 against 0.811 to 0.848 for the random crop, and a single-split run with a different mask reversed that ordering, so we make no localization claim.

Patient state. Patient infection status was predictable from uninfected cells alone at patient-level AUC 0.946 to 0.954, and a probe identified the correct patient among 201 at top-1 accuracy 0.50 to 0.58 (chance 0.005, Appendix H). These results fit a slide- or patient-level cue, which unavailable slide metadata prevent us from identifying.

Limits and conclusion. Ablation and erasure act on frozen representations and do not identify causes, and the TB extent verdict rests mainly on xrv features, where calibration shows lower power (Appendix M). Each domain's first verdict led to a different next test. The TB failure fits a source effect that population differences also explain, the glioma refit signal depends on preprocessing, and the malaria signal persists after parasite-pixel replacement. The TB failure cause, the malaria cue and the non-definitional glioma signal are unresolved.

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Appendix A. Statements

Data and Code Availability. All data are public and de-identified: Shenzhen and Montgomery chest radiographs (Jaeger et al., 2014), the Belarus TB Portals collection (Rosenthal et al., 2017) (Kaggle mirror), RSNA Pneumonia Challenge normals (Shih et al., 2019), the TCGA LGG/GBM cohort (Ceccarelli et al., 2016) via cBioPortal, and the NIH NLM malaria cell images (Rajaraman et al., 2018). No new data were collected. Code and result files are anonymized for review and a repository link will be added on acceptance.

IRB. Secondary analysis of public de-identified datasets, not human subjects research requiring IRB approval.

Author contributions. Disclosed at the camera-ready stage, per ML4H anonymization requirements.

Appendix B. Audit Workflow Figure

Figure 2 summarizes the audit described in Section 3.

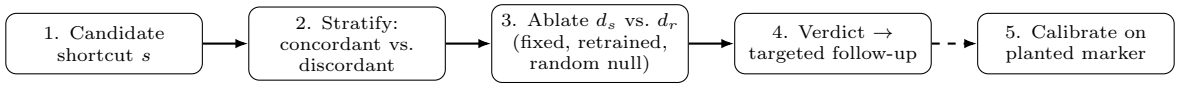


Figure 2: The audit. Step 3 reports every ablation with a retrained head and a random-direction null, step 4 selects the follow-up for each domain (Section 5), and step 5 calibrates the audit itself.

Appendix C. Literature Gaps, Contributions and Novelty Scope

Search. Searches were run in October 2026 with a web search tool, followed by reading the arXiv or journal abstract pages of the closest hits. Query themes were shortcut testing in medical AI, planted or synthetic shortcut calibration of audits, concept erasure (LEACE, INLP) in medical imaging, iterative or multi-shortcut auditing with retrained-head or random-direction controls, shortcut and source confounds in TB radiograph classifiers, cross-dataset generalization on Shenzhen and Montgomery, patient- or slide-level leakage in the NIH malaria cell dataset, grade confounding in IDH prediction, and representational similarity (CKA) after removing a decodable concept. Only abstracts were read for several 2026 preprints, so details beyond their abstracts may differ. The search was not a systematic review, so every statement below applies only to what it found and must be rechecked before camera-ready.

Gaps in the literature. In this search we found the following gaps, each stated against the closest work.

G1. *No calibration of medical-imaging shortcut audits.* Shortcut tests, probes and counterfactual edits exist for chest radiographs (Jiménez-Sánchez et al., 2022; Pedersen et al., 2026; Cooke et al., 2025; Brown et al., 2023), and semi-synthetic testbeds exist for explanation methods (Adebayo et al., 2022). We found none that reports the false-flag rate and detection power of a representation-level audit as a function of the strength of a planted shortcut in medical images, per encoder.

G2. *No check of linear erasure with nonlinear probes for a label-coincident medical-imaging source confound.* Site or source signal has been removed by null-space projection in brain-MRI embeddings (Rahbar, 2026), and ablating a probe direction is known not to test necessity in language-model interpretability (Jiang, 2026). We found no radiograph study that applies LEACE and INLP to a source confound that coincides with the label and then probes the residual with nonlinear models.

G3. *No eraser transfer across encoders in medical imaging.* We found no study that fits an eraser in one frozen medical-imaging encoder and tests it in another.

G4. *No source-matched evaluation of a Belarus TB Portals plus RSNA detector.* Dataset specificity of TB detectors is known (Sathitratanacheewin and Pongpirul, 2018), and source identification has been studied for other chest radiograph sets (Dack and Dai, 2025). We found no evaluation of a detector built from Belarus TB Portals cases (Rosenthal et al., 2017) and RSNA normals (Shih et al., 2019) on data where both classes share a source.

G5. *No patient-state test beyond slide-disjoint splits in NIH malaria.* Slide-disjoint evaluation lowers reported accuracy (Hou et al., 2026). We found no test of whether uninfected cells carry patient infection status, or of parasite-removed views under patient-grouped splits.

G6. *No shortcut audit of grade in tabular IDH prediction.* The WHO classification defines IDH-based entities (Louis et al., 2021). We found no study that audits grade as a shortcut for IDH prediction on TCGA and reports that ablating it collapses to a constant.

G7. *No CKA check after erasing a known medical confound.* CKA is interpreted through linear decoders (Harvey et al., 2024) and has known reliability problems (Davari et al., 2023). We found no study that reports it before and after erasing a known imaging confound.

G8. *No functional-use control combined with a planted shortcut.* Restorability of shortcuts after unlearning is tested on natural-image and dermatology benchmarks (Lu and Ji, 2026), and calibration shifts under shortcuts are studied for chest drains (Kina and Petersen, 2026). We found no audit that combines fixed head, retrained head and random-direction nulls for every ablation and validates it against a planted shortcut.

Contributions. To our knowledge, this is the first study to do the following, and each statement is limited to the gaps above.

C1. Report the false-flag rate (0.00 at $\rho = 0.5$ for three encoders) and detection power (flag rate 0.85 to 1.00 at $\rho \geq 0.7$ for DINOv2 and ResNet-50, and 0.30 to 0.75 for xrv at $\rho = 0.7$ to 0.9) of a representation-level shortcut audit against a planted marker in medical images, and to show that the audit can miss the strongest planted shortcut (xrv at $\rho = 1.0$).

C2. Evaluate a TB detector built from Belarus TB Portals cases and RSNA normals on source-matched Shenzhen and Montgomery data, where it scored AUC 0.455 to 0.542 although within-source detection was 0.893 to 0.903, and to show that format harmonization lifted it to 0.752 only partly.

C3. Show on a medical-imaging source confound that coincides with the label that linear erasure (LEACE, 200-round INLP) leaves source decodable by MLP (0.941 to 0.998) and GBM probes, with matched-data detection unchanged.

C4. Test eraser transport and head inheritance across frozen medical-imaging encoders, and to report that source decodability by an MLP stays at 0.991 or above under every transported eraser (Appendix J).

C5. Report CKA before and after erasing a known imaging confound: CKA with the original encoder stays at 0.890 to 0.978 while linear source decodability falls to chance.

C6. Test, in NIH malaria cell images, whether uninfected cells from patients with an infected cell differ from uninfected cells of patients with none (AUC 0.612 to 0.675) and whether patient infection status can be predicted from uninfected cells (AUC 0.946 to 0.954).

C7. Report that ablating WHO grade in TCGA tabular IDH prediction collapses predictions to a constant, and that the grade-excluded refit ranges from 0.563 to 0.812 across imputation choices.

C8. Methodological: an audit workflow that reports every ablation with fixed, retrained and random-direction controls, and an explicit record of deviations and non-reproduced results (Appendix L). We do not claim this record as a first.

What we do not claim. We do not claim that shortcuts in TB, malaria or glioma data are new findings in general, that direction ablation, probing, LEACE, INLP or CKA are new, that planting a marker is new, that slide leakage in NIH malaria is new (Hou et al., 2026), or that the audit establishes causes. We do not claim that the TB source difference is format alone, since populations also differ. We do not claim that malaria staining is irrelevant or that any malaria signal is localized.

Table 1 lists each contribution next to its closest precedent. The ingredients all have precedent. What we did not find is their combination in one audit that is calibrated against a planted marker, checked with a retrained head and random-direction null for every ablation, and applied to a label-coincident source confound with nonlinear probes.

Table 1: Contributions, closest precedent found, and the scope of our novelty statement.

Contribution	Closest precedent found	Difference
C1 Calibrated audit	Semi-synthetic testbeds for explanations (Adebayo et al., 2022), counterfactual edits (Cooke et al., 2025) and layer-wise probes (Pedersen et al., 2026)	Operating characteristics (false flags, power by ρ) of a representation-level audit, per encoder, in medical images.
C2 TB source-matched test	Dataset specificity (Sathitratanacheewin and Pongpirul, 2018), non-lung cues (Maguolo and Nanni, 2020) and site effects (Zech et al., 2018)	Detector built from Belarus TB Portals and RSNA, tested where both classes share a source.
C3 Erasure and nonlinear probes	INLP (Ravfogel et al., 2020), LEACE (Belrose et al., 2023), site projection in brain MRI (Rahbar, 2026) and probe refitting (Jiang, 2026)	Label-coincident confound in radiographs, LEACE and INLP followed by MLP and GBM probes. We do not claim a new eraser.
C4 Transport	None found in medical imaging	Eraser transport through a ridge map and stitched heads, with random rank-1 control.
C5 CKA	CKA as linear decoding (Harvey et al., 2024) and its reliability (Davari et al., 2023)	Before/after erasing a known confound, with random rank-1 control.
C6 Malaria patient state	Slide-disjoint evaluation (Hou et al., 2026)	Parasite-removed views, bystander test, patient identifiability. We do not identify the cue.
C7 Glioma grade	IDH-based classification (Louis et al., 2021)	Constant-prediction collapse and refit sensitivity. Shows that the grade ablation carries no subgroup information.
C8 Controls and record	Last-layer retraining (Kirichenko et al., 2023), shortcut testing (Brown et al., 2023) and restorability (Lu and Ji, 2026)	Combined controls for every ablation, and a record of deviations.

Appendix D. Protocol Details

Data. *TB.* Belarus TB Portals DICOM cases (517 pulmonary-only, 533 with extrapulmonary disease) and 8,851 RSNA normals form the training population. Shenzhen and Montgomery radiographs, in which TB and normals share an acquisition pipeline, form the source-matched evaluation. *Glioma.* Tabular TCGA data for 995 patients with IDH status, with WHO grade, MGMT, 1p/19q and TERT as inputs (11.6%, 2.1% and 67% missing for the last three). The concordant test set has 257 patients and the discordant set 141 (104 IDH-wildtype low-grade, 37 IDH-mutant glioblastoma). *Malaria.* NIH thin-smear cell images with 201 patient groups parsed from file names, median 85 cells per group, 151 groups with at least one infected cell, 10,364 uninfected cells from those patients and 3,415 from the others.

Features. Frozen encoders: torchxrayvision DenseNet-121 (xrv, 1024-d) (Cohen et al., 2022), DINOv2-small and ResNet-50 for TB, and ResNet-50, DenseNet-121 and DINOv2-small for malaria. Glioma uses the input variables directly. Harmonization applies histogram equalization followed by a JPEG quality-90 round trip to all TB sources before feature extraction.

Classifiers and probes. Heads and shortcut probes are logistic regressions. Source probes after erasure use logistic regression, an MLP with one hidden layer of 256 units and early stopping, and a histogram gradient-boosting classifier with 150 iterations. Erasers are fit on 60% of the Belarus/RSNA images and probes are tested on the remaining 40%. INLP refits a logistic probe and projects onto its null space each round, for up to 200 rounds. LEACE is the closed-form binary-concept eraser. Zero-variance feature columns are excluded from the fit and passed through, because including them let the erased direction leak back into those columns in the 1024-d xrv features and left a linear probe at AUC 0.997 before this fix. In our diagnostic, 108 of the 1024 xrv columns had zero variance (covariance rank 916) and the largest standardized test value was 931 standard deviations, and LEACE on the full space left a linear probe at train AUC 1.000 although all class means were equal to 10^{-11}. Dropping the dead columns gave train 0.509 and test 0.524, and clipping outliers made no difference. Our best explanation, supported by this diagnostic but not independently proven, is that the erased direction leaks into the dead columns at floating-point level and is re-amplified by the probe's standardization. The planted-marker runs apply LEACE after a 64-d PCA and were never affected.

Ablation. For a unit direction $\hat{d}$ the ablated feature is $z - (z^\top \hat{d})\hat{d}$. The shortcut direction d_s is the weight vector of a logistic probe for s, the classifier direction d_c is the weight vector of the logistic classifier, and the residual direction d_r is d_c minus its projection on d_s, normalized. Each ablation is evaluated with the original head (fixed), with a head retrained on ablated features, and against 200 covariance-matched random directions, from which an empirical p-value is computed. The number of seeds is 3 for TB xrv, 2 for TB DINOv2, 5 for malaria and 10 for glioma.

Malaria shortcut and views. The staining shortcut is a high-versus-low mean saturation split. The parasite region keeps dilated mask pixels, cropped around the centre. The RBC view replaces dilated mask pixels by the median colour of the remaining cell pixels. The random crop is a circle of radius 35% of the smaller image side, off-centre, chosen from up to 50 candidates for least overlap with the dilated mask (mean overlap 1.2%, 28.95% of crops with any overlap). Grayscale is standardized within the cell, and the silhouette keeps only the cell outline. Handcrafted features are colour statistics and size-and-shape statistics. Masks are approximate thresholds (a fixed HSV rule or a per-cell adaptive rule, depending on the encoder run) and were not validated against annotations. A coarser P# grouping with ResNet-50 gave the same patient-grouped AUCs within 0.001 to 0.002 for every view.

Planted marker. A bright corner block with a soft gradient is added to Shenzhen and Montgomery radiographs with $P(\text{marker} \mid \text{TB}) = \rho$, so ρ is the marker prevalence among TB cases and $\rho = 0.5$ is the no-association setting. Features are reduced to a 64-d PCA space and the procedure is repeated over 20 splits per ρ. The audit flags reliance when LEACE on the recovered marker direction raises AUC on marker-flipped test images by more than 0.02 and above the 95th percentile of the random-direction null gain. The shortcut tax is concordant AUC minus flipped AUC.

Statistics. Intervals are 95% percentile bootstrap intervals over test images, or over patients in the patient-level tests, and exclude training variance. Intervals for malaria view AUCs resample cells and ignore patient clustering, so they are too narrow. The RBC-versus-random-crop comparison is a paired bootstrap against an equivalence margin of ± 0.02.

Separate-pipeline items. The TB 3×3 grid (Table 6), the sensitivity gap, the extent-only Belarus DICOM arm and a single-split malaria ResNet-50 run (RBC-only AUC 0.891, random crop 0.907, full image 0.976) come from a separate analysis pipeline and were not recomputed under the protocol above.

Appendix E. Index of Controls

Table 2 maps every control to where it is reported, so a reader can find the full result behind each statement in the main text.

Table 2: Controls and where they are reported.

Control	Location
Retrained head, random-direction null	Table 5
Planted-marker calibration	Section 5.1, Appendix I
Source-matched TB transfer, metadata-only baseline	Table 3
Linear, MLP and GBM probes after erasure	Table 4, Figure 3a
Harmonized TB variant	Tables 3 and 4
Eraser transport, head inheritance, CKA	Tables 17, 18 and 19
Supplementary analyses from the separate pipeline	Appendix K
Glioma refits, imputation variants, label-permutation null	Tables 7 and 8
Patient-grouped malaria splits, three encoders	Table 10
Parasite-removed views, mask overlap	Table 10, Appendix D
Bystander and patient-identifiability tests	Tables 11, 13, 14 and 12

Appendix F. TB: Extended Results

Source-matched transfer. Within-source detection is strong (Shenzhen 0.893 to 0.903 and Montgomery 0.844 to 0.903 across the three unharmonized encoders, Table 3). Transfer between the matched sources is partial. Shenzhen to Montgomery gives 0.628 to 0.736, and Montgomery to Shenzhen gives 0.787 (DINOv2) and 0.821 (ResNet-50) but 0.356 for xrv, below chance, which we cannot explain. The Belarus/RSNA-trained detector scores 0.395 (DINOv2), 0.465 (ResNet-50) and 0.501 (xrv) on Shenzhen alone, and 0.569, 0.690 and 0.673 on Montgomery alone, so its failure is concentrated on Shenzhen. The cosine between the source direction and the direction of a detector trained on matched data is 0.045, 0.031 and 0.002 for DINOv2, ResNet-50 and xrv, so the source direction is nearly orthogonal to the direction the matched detector uses.

Metadata and harmonization. Image size and intensity statistics alone separate Belarus from RSNA at AUC 1.000 and predict TB within the matched sets at 0.748 [0.713, 0.780], so image-level statistics separate TB from normal even within the matched sets. Harmonization leaves source decodable (0.999) and raises the Belarus/RSNA-trained detector on matched data from 0.542 to 0.752 for xrv, 0.802 on Shenzhen and 0.705 on Montgomery. Because harmonization changes format while population differences remain, this shows format contributes to the failure and does not show that format is the only cause.

Erasure. LEACE reaches linear chance in one step (0.470 to 0.494 for the four feature sets). INLP needs 100 rounds for DINOv2 (0.496) and 200 for ResNet-50 (0.484), xrv (0.501) and harmonized xrv (0.503) (Figure 3a, Table 4). After 200 rounds MLP probes still reach 0.941 (DINOv2) and at least 0.990 for the others, and GBM probes reach 0.831 (DINOv2) and at least 0.991 for the others. Matched-data TB detection is unchanged by erasure (None versus LEACE: DINOv2 0.839 vs. 0.844, ResNet-50 0.850 vs. 0.847, xrv 0.808 vs. 0.825, harmonized 0.812 vs. 0.827), so removing the linear source direction does not remove information that the matched detector needs.

Extent ablation. In TB xrv features, ablating the extent direction leaves positive-class accuracy near baseline but lowers normal accuracy from 0.998 to 0.835 ($p = 0.01$), and a retrained head restores 0.998 (Table 5). In DINOv2 features the same ablation changes nothing (overall 0.999 to 0.998, $p = 0.92$). The residual ablation with a fixed head zeroes positive-class accuracy in DINOv2 (0.000 for both groups) and lowers it to 0.108 and 0.236 in xrv, while retrained heads return to 0.976 to 0.989, so it removes weights without removing information. Table 6 gives the train/test grid of the separate pipeline.

Table 3: TB source-matched evaluation (AUC with 95% bootstrap interval over test images). Metadata (image size and intensity statistics) alone separate Belarus from RSNA at AUC 1.000 and predict TB within Shenzhen and Montgomery at AUC 0.748 [0.713, 0.780].

Test	DINOv2	ResNet-50	xrv	xrv harmonized
Source probe, Belarus vs. RSNA	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.999 [0.999, 1.000]	0.999 [0.999, 1.000]
Within Shenzhen (CV)	0.893 [0.869, 0.916]	0.897 [0.873, 0.920]	0.903 [0.879, 0.925]	0.930 [0.910, 0.948]
Within Montgomery (CV)	0.844 [0.775, 0.906]	0.870 [0.809, 0.922]	0.903 [0.840, 0.954]	0.898 [0.829, 0.951]
Shenzhen → Montgomery	0.736 [0.652, 0.820]	0.718 [0.624, 0.802]	0.628 [0.528, 0.722]	0.548 [0.446, 0.645]
Montgomery → Shenzhen	0.787 [0.753, 0.820]	0.821 [0.791, 0.851]	0.356 [0.318, 0.398]	0.608 [0.567, 0.650]
Belarus/RSNA → Shenzhen+Montgomery	0.455 [0.417, 0.499]	0.511 [0.471, 0.552]	0.542 [0.503, 0.584]	0.752 [0.715, 0.785]
Belarus/RSNA → Shenzhen	0.395 [0.350, 0.436]	0.465 [0.418, 0.507]	0.501 [0.455, 0.545]	0.802 [0.767, 0.835]
Belarus/RSNA → Montgomery	0.569 [0.466, 0.670]	0.690 [0.598, 0.779]	0.673 [0.573, 0.762]	0.705 [0.609, 0.792]
Cosine(source dir., matched detector dir.)	0.045	0.031	0.002	0.001

Table 4: Held-out source AUC after erasure (linear / MLP / GBM probes) and matched-data TB detection AUC (cross-validated on Shenzhen+Montgomery, "det."). Erasers are fit on 60% of the Belarus/RSNA images and probes are tested on the remaining 40%. n/a means the quantity was not computed.

	DINOv2				ResNet-50				xrv				xrv harmonized			
Eraser	lin.	MLP	GBM	det.	lin.	MLP	GBM	det.	lin.	MLP	GBM	det.	lin.	MLP	GBM	det.
None	1.000	1.000	1.000	0.839	1.000	1.000	0.999	0.850	0.999	0.999	1.000	0.808	0.999	0.999	1.000	0.812
LEACE	0.494	0.998	0.985	0.844	0.470	0.997	0.995	0.847	0.484	0.989	1.000	0.825	0.481	0.998	1.000	0.827
INLP 1	0.999	0.999	0.999	n/a	0.999	1.000	0.999	n/a	0.999	0.994	0.990	n/a	0.999	0.999	0.992	n/a
INLP 10	0.967	0.998	0.990	n/a	0.994	0.999	0.999	n/a	0.988	0.991	1.000	n/a	0.988	0.999	1.000	n/a
INLP 50	0.671	0.996	0.977	0.831	0.905	0.999	0.998	0.854	0.882	0.996	1.000	0.825	0.901	0.999	1.000	0.831
INLP 100	0.496	0.989	0.953	n/a	0.788	0.999	0.997	n/a	0.684	0.999	1.000	n/a	0.658	0.999	0.999	n/a
INLP 200	0.472	0.941	0.831	n/a	0.484	0.994	0.991	n/a	0.501	0.990	0.999	n/a	0.503	0.991	0.995	n/a

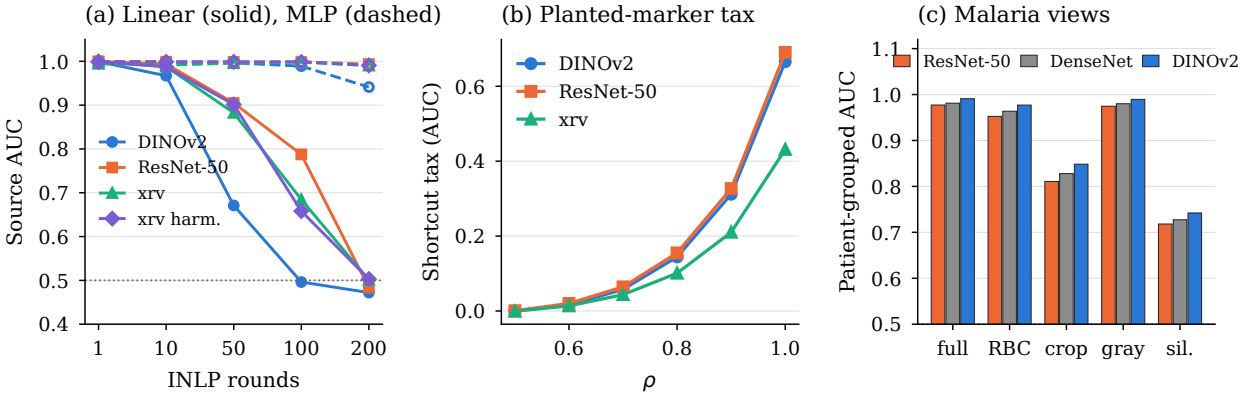


Figure 3: (a) Source AUC after INLP by number of rounds (linear probe solid, MLP probe dashed), for four feature sets. (b) Planted-marker shortcut tax (concordant minus flipped AUC) by ρ. (c) Malaria patient-grouped AUC by view and encoder (chance 0.5).

Table 5: Ablation controls, accuracy (empirical p against covariance-matched random directions in parentheses). Seeds: TB xrv 3, TB DINOv2 2, malaria 5, glioma 10. Malaria shortcut is high vs. low mean saturation. Single-class cells carry little information on their own. TB uses the extent direction fit within TB-positive cases.

Domain	Cell	Baseline	Shortcut, fixed head	Shortcut, retrained	Residual, fixed head	Residual, retrained
TB xrv	Group A positives	0.991	0.991 (0.79)	0.989 (1.00)	0.108 (0.00)	0.989 (1.00)
TB xrv	Group B positives	0.986	0.990 (0.56)	0.988 (1.00)	0.236 (0.00)	0.986 (0.74)
TB xrv	Normals	0.998	0.835 (0.01)	0.998 (1.00)	0.290 (0.00)	0.998 (0.74)
TB xrv	All	0.997	0.851 (0.01)	0.997 (0.97)	0.278 (0.00)	0.997 (1.00)
TB DINOv2	Group A positives	1.000	1.000 (1.00)	1.000 (1.00)	0.000 (0.00)	0.986 (0.05)
TB DINOv2	Group B positives	0.991	0.991 (1.00)	0.991 (1.00)	0.000 (0.00)	0.976 (0.10)
TB DINOv2	Normals	0.999	0.999 (0.79)	0.999 (1.00)	1.000 (0.60)	0.997 (0.03)
TB DINOv2	All	0.999	0.998 (0.92)	0.999 (1.00)	0.894 (0.02)	0.995 (0.03)
Malaria full	All	0.928	0.927 (0.95)	0.928 (0.45)	0.506 (0.01)	0.931 (0.03)
Malaria full	Parasitized	0.922	0.922 (0.95)	0.923 (0.65)	0.200 (0.00)	0.924 (0.03)
Malaria full	Uninfected	0.934	0.934 (0.93)	0.935 (0.58)	0.800 (0.38)	0.939 (0.03)
Malaria RBC	All	0.890	0.886 (0.81)	0.890 (0.74)	0.504 (0.00)	0.893 (0.03)
Malaria RBC	Parasitized	0.879	0.880 (0.93)	0.879 (0.87)	0.328 (0.00)	0.879 (0.03)
Malaria RBC	Uninfected	0.902	0.893 (0.91)	0.902 (0.84)	0.703 (0.28)	0.908 (0.06)
Glioma	Concordant	1.000	0.490 (0.12)	0.815 (0.11)	0.490 (0.12)	0.815 (0.11)
Glioma	Discordant IDH-wt LGG	0.000	0.000 (1.00)	0.654 (0.06)	0.000 (1.00)	0.654 (0.06)
Glioma	Discordant IDH-mut GBM	0.000	1.000 (0.06)	0.622 (0.03)	1.000 (0.06)	0.622 (0.03)

Table 6: TB train/test grid from the original run (8,851 RSNA normals). A and B are pulmonary-only and extrapulmonary groups. All cells meet the WHO TPP.

Train	Test	Sens.	Spec.	AUC
A	A	0.977	0.996	0.993
A	B	0.970	0.996	0.995
A	A+B	0.970	0.996	0.996
B	A	0.953	0.995	0.996
B	B	0.955	0.995	0.998
B	A+B	0.970	0.995	0.998
A+B	A	0.992	0.995	0.997
A+B	B	1.000	0.995	0.999
A+B	A+B	0.973	0.995	0.998

Appendix G. Glioma: Extended Results

Grade ablation. The grade direction has cosine 1.00 with the classifier direction (Table 9), so ablating it removes the classifier's only informative input. The ablated classifier predicts IDH-mutant for all 398 test patients. Accuracy on the concordant set (0.490) is below the 0.510 obtained by predicting all-wildtype, and the apparent gain on IDH-mutant glioblastoma (1.000) is the all-mutant prior. The fixed-head and retrained-head accuracies for the three subgroups are in Table 5, where a retrained head reaches 0.815, 0.654 and 0.622 on the concordant, IDH-wildtype low-grade and IDH-mutant glioblastoma groups.

Refits without grade. Table 7 lists every variant. The refit with zero imputation reaches discordant AUC 0.738, median imputation 0.812, and median imputation with missingness flags 0.563 with a wide interval [0.394, 0.596], so the estimate depends on analyst choices. Dropping 1p/19q lowers it to 0.624 and dropping 1p/19q and TERT gives 0.638, identical to the MGMT-only model since MGMT is then the only input. TERT alone is below chance on the discordant set (0.336), and 1p/19q alone is 0.527. The label-permutation null means are 0.422 to 0.528. Missingness indicators alone predict IDH at 0.666 [0.631, 0.699] (Table 8), so part of any refit signal can come from which assays were run and not from their values.

Stratified by grade. With grade held constant within a stratum, the refit reaches 0.891 [0.855, 0.925] in low-grade glioma and 0.606 [0.509, 0.704] in glioblastoma, and without 1p/19q and TERT 0.750 and 0.604. The within-glioblastoma signal is therefore weak. Because 1p/19q co-deletion implies IDH mutation by definition (Louis et al., 2021), the within-low-grade value is partly definitional.

Table 7: Glioma refits (20 splits). Discordant AUC with 95% bootstrap interval, and mean of the label-permutation null.

Variant	Conc. AUC	Disc. AUC	Perm. null
full (with grade)	1.000	0.000 [0.000, 0.000]	0.476
no grade [paper refit]	0.877	0.738 [0.649, 0.823]	0.452
no grade, median-impute	0.806	0.812 [0.736, 0.888]	0.428
no grade, median+missing flags	0.948	0.563 [0.394, 0.596]	0.422
no grade, no 1p/19q	0.834	0.624 [0.544, 0.716]	0.464
no grade, no 1p/19q, no TERT	0.809	0.638 [0.548, 0.723]	0.528
1p/19q only	0.698	0.527 [0.500, 0.566]	0.500
TERT only	0.581	0.336 [0.284, 0.387]	0.484
MGMT only	0.809	0.638 [0.548, 0.723]	0.528

Table 8: Glioma robustness. Cross-validated AUC with 95% bootstrap interval for grade-stratified refits and for missingness indicators alone.

Setting	AUC [95% CI]
no grade [paper refit] | LGG only	0.891 [0.855, 0.925]
no grade [paper refit] | GBM only	0.606 [0.509, 0.704]
no grade, no 1p/19q, no TERT | LGG only	0.750 [0.686, 0.808]
no grade, no 1p/19q, no TERT | GBM only	0.604 [0.510, 0.703]
Missingness indicators only	0.666 [0.631, 0.699]

Table 9: Glioma grade ablation: fraction predicted IDH-mutant (accuracy in parentheses) per test subgroup. The ablated classifier predicts IDH-mutant for everyone. Majority-class accuracy on the concordant set is 0.510. Cosine between the grade direction and the classifier direction: 1.00.

Subgroup	n	Baseline	Grade ablated
Concordant	257	0.490 (1.000)	1.000 (0.490)
Disc. IDH-wt LGG	104	1.000 (0.000)	1.000 (0.000)
Disc. IDH-mut GBM	37	0.000 (0.000)	1.000 (1.000)

Appendix H. Malaria: Extended Results

Splitting. Patient-grouped AUC is 0.002 to 0.005 below cell-level AUC for the full image and 0.002 to 0.033 below it across all views (Table 10). In total, 24.9% of patient groups, holding 12.4% of cells, contain a single class. These groups offer no within-patient contrast, so for them patient identity and label coincide.

Views. Across the three encoders the full image reaches 0.977 to 0.991, grayscale 0.975 to 0.989, the RBC view 0.952 to 0.977, the parasite region alone 0.899 to 0.928, the random crop 0.811 to 0.848, and the silhouette 0.718 to 0.742 (Figure 3c). Size-and-shape features alone reach 0.575 to 0.577, colour-only features 0.967 to 0.969, and all handcrafted features except the parasite fraction 0.971 to 0.972. The RBC view minus the random crop is 0.142 for ResNet-50, 0.136 for DenseNet-121 and 0.129 for DINOv2, each with an interval that excludes the ± 0.02 equivalence margin. With a mean mask overlap of 1.2% the random crop contains little parasite, so the gap is not caused by the crop containing parasite pixels. It could reflect crop size or position, so it is not an estimate of how much information lies outside the parasite.

Staining ablation. A saturation-defined staining direction removed with a fixed head leaves ResNet-50 accuracy at 0.927 against 0.928 at baseline ($p = 0.95$), and in the RBC view at 0.886 against 0.890 ($p = 0.81$) (Table 5). The residual ablation with a fixed head reduces accuracy to 0.506 and 0.504 and a retrained head returns it to 0.931 and 0.893. A direction defined by saturation is one proxy for staining and excludes other stain properties, so the null result does not establish that stain is irrelevant.

Patient state. Uninfected cells from patients with at least one infected cell (10,364 cells from 151 patients) have higher infected scores than uninfected cells from the 50 patients with none (3,415 cells), with AUC 0.612 to 0.675 for full-image features and 0.505 to 0.512 for the parasite region (Table 11). In ResNet-50 the false-positive rate is 0.068 for bystander cells and 0.040 for healthy-patient cells (Table 13). Patient infection status can be predicted from uninfected cells alone at patient-level AUC 0.946 to 0.954 for full-image features, and a probe identifies the source patient among 201 at top-1 accuracy 0.499 to 0.582 for full images and 0.491 to 0.582 for the RBC view, against a chance of 0.005 (Table 12). The silhouette alone identifies the patient at 0.068 to 0.098, more than ten times chance. In ResNet-50 the random-crop and silhouette views give larger bystander excess false positives than the full image (0.246 vs. 0.111 and 0.364 vs. 0.187, Table 14), so the patient-state cue is not confined to parasite pixels. These results are compatible with slide-level or patient-level appearance, which cell-level splitting would leak, but slide metadata are unavailable, so the cue is not identified.

Table 10: Malaria, patient-grouped (5-fold, 201 groups) and cell-level cross-validated AUC with 95% intervals. RBC view minus random crop (patient-grouped, paired bootstrap): ResNet-50 0.142 [0.137, 0.146], DenseNet-121 0.136 [0.131, 0.140], DINOv2 0.129 [0.124, 0.133]. The ± 0.02 equivalence test failed for all three.

View	Patient-grouped			Cell-level		
	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2
Full image	0.977 [0.975, 0.978]	0.981 [0.980, 0.982]	0.991 [0.990, 0.992]	0.982 [0.980, 0.983]	0.986 [0.985, 0.987]	0.993 [0.992, 0.993]
Parasite region	0.899 [0.895, 0.904]	0.917 [0.914, 0.921]	0.928 [0.925, 0.931]	0.907 [0.903, 0.911]	0.926 [0.923, 0.929]	0.936 [0.934, 0.940]
RBC view	0.952 [0.950, 0.954]	0.964 [0.962, 0.966]	0.977 [0.975, 0.978]	0.964 [0.963, 0.966]	0.974 [0.972, 0.975]	0.982 [0.980, 0.983]
Random crop	0.811 [0.805, 0.815]	0.828 [0.823, 0.833]	0.848 [0.844, 0.852]	0.844 [0.839, 0.849]	0.860 [0.855, 0.864]	0.874 [0.870, 0.877]
Grayscale	0.975 [0.973, 0.976]	0.980 [0.979, 0.981]	0.989 [0.988, 0.990]	0.978 [0.977, 0.980]	0.984 [0.983, 0.985]	0.991 [0.990, 0.992]
Silhouette	0.718 [0.713, 0.723]	0.727 [0.721, 0.732]	0.742 [0.737, 0.747]	0.726 [0.721, 0.731]	0.737 [0.731, 0.743]	0.751 [0.745, 0.756]
Colour-only features	0.968 [0.966, 0.970]	0.969 [0.967, 0.970]	0.967 [0.965, 0.969]	0.982 [0.981, 0.984]	0.983 [0.982, 0.984]	0.983 [0.982, 0.984]
Size and shape only	0.575 [0.568, 0.582]	0.576 [0.569, 0.583]	0.577 [0.570, 0.583]	0.586 [0.579, 0.593]	0.585 [0.578, 0.591]	0.588 [0.581, 0.595]

Table 11: Malaria patient-state test. (i) AUC separating uninfected cells of patients with at least one infected cell (10,364 cells) from uninfected cells of patients with none (3,415 cells), using the infected-vs.-uninfected score of a patient-held-out classifier. (ii) Patient-level AUC for predicting whether a patient has any infected cell from uninfected cells alone (scores averaged per patient, 95% cluster bootstrap over patients).

	(i) Bystander vs. healthy-patient AUC			(ii) Patient-level status AUC		
View	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2
Full	0.644	0.675	0.612	0.954 [0.909, 0.987]	0.949 [0.904, 0.982]	0.946 [0.903, 0.981]
Parasite	0.505	0.505	0.512	0.796 [0.721, 0.857]	0.875 [0.821, 0.922]	0.886 [0.825, 0.938]
RBC	0.669	0.678	0.651	0.953 [0.910, 0.988]	0.948 [0.902, 0.980]	0.949 [0.904, 0.981]
Random crop	0.665	0.690	0.699	0.911 [0.861, 0.954]	0.930 [0.881, 0.969]	0.941 [0.894, 0.977]
Gray	0.568	0.592	0.608	0.934 [0.883, 0.967]	0.948 [0.909, 0.978]	0.909 [0.846, 0.952]
Silhouette	0.645	0.662	0.660	0.886 [0.830, 0.935]	0.874 [0.812, 0.928]	0.854 [0.788, 0.909]

Table 12: Malaria patient-identifiability probe: top-1 accuracy of identifying the source patient among 201 from a held-out cell (chance 0.005).

View	ResNet-50	DenseNet-121	DINOv2
Full image	0.499	0.582	0.567
RBC view	0.491	0.582	0.567
Random crop	0.330	0.367	0.428
Grayscale	0.336	0.402	0.395
Silhouette	0.068	0.074	0.098

Table 13: Malaria bystander test, full-image features. False-positive rate of the infected-vs.-uninfected classifier on uninfected cells from patients with (bystander) and without (healthy) an infected cell, and cell-level AUC of predicting patient status from uninfected cells. 201 patients, 151 with an infected cell.

Quantity	ResNet-50	DenseNet-121	DINOv2
FPR, bystander	0.068	0.056	0.040
FPR, healthy patient	0.040	0.034	0.028
Cell-level status AUC	0.885	0.900	0.903

Table 14: Malaria bystander test for every view: false-positive rate of the infected-vs.-uninfected classifier on uninfected cells from patients with an infected cell (bystander) and from patients with none (healthy), and cell-level AUC of predicting patient status from uninfected cells.

	FPR bystander			FPR healthy patient			Cell-level status AUC		
View	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2
Full	0.068	0.056	0.040	0.040	0.034	0.028	0.885	0.900	0.903
Parasite	0.075	0.067	0.059	0.073	0.060	0.050	0.537	0.556	0.590
RBC	0.108	0.083	0.073	0.052	0.044	0.039	0.886	0.900	0.905
Random crop	0.246	0.227	0.192	0.111	0.097	0.088	0.837	0.874	0.895
Gray	0.079	0.060	0.044	0.048	0.043	0.029	0.833	0.869	0.842
Silhouette	0.364	0.364	0.352	0.187	0.164	0.158	0.714	0.741	0.739

Appendix I. Planted Marker: Extended Results

The flag rate is 0.00 at $\rho = 0.5$ for all three encoders (Table 15). For DINOv2 it is 0.40, 0.85, 1.00, 1.00 and 0.90 for $\rho = 0.6$ to 1.0, for ResNet-50 0.25, 0.85, 1.00, 1.00 and 1.00, and for xrv 0.00, 0.30, 0.75, 0.70 and 0.00. The shortcut tax rises with ρ for all encoders (Figure 3b) and is smaller for xrv (0.431 at $\rho = 1.0$ against 0.666 and 0.691). The recovered direction has cosine 0.64 to 0.69 with the planted direction for DINOv2 and 0.66 to 0.72 for ResNet-50, but 0.25 to 0.27 for xrv, so the recovered xrv direction is only weakly aligned with the marker.

At $\rho = 1.0$ the marker is the only label-correlated signal, so a retrained head cannot recover (retrained flipped AUC 0.517 for DINOv2, 0.507 for ResNet-50 and 0.540 for xrv). For xrv the LEACE gain at $\rho = 1.0$ is negative (-0.182, Table 16), which is why the flag rate falls to 0.00 there, and for DINOv2 it is 0.142 with flag rate 0.90. A flag at $\rho = 1.0$ is therefore not guaranteed, and a clean audit at very high marker correlation should not be read as absence of the marker. After LEACE the marker presence AUC from the residual is near 0.5 (0.510 for ResNet-50 at $\rho = 0.7$, Table 16), so the eraser removes the linearly decodable marker.

Table 15: Planted-marker calibration (20 splits per ρ): flag rate, shortcut tax (concordant minus flipped AUC), cosine between recovered and planted direction (in a 64-d PCA space), and AUC on flipped images after erasing the marker and retraining the head.

	DINOv2				ResNet-50				xrv			
ρ	flag	tax	cos	retr.	flag	tax	cos	retr.	flag	tax	cos	retr.
0.5	0.00	0.001	0.68	0.867	0.00	0.001	0.72	0.881	0.00	-0.001	0.26	0.895
0.6	0.40	0.016	0.69	0.869	0.25	0.020	0.72	0.882	0.00	0.014	0.27	0.887
0.7	0.85	0.058	0.69	0.878	0.85	0.064	0.71	0.890	0.30	0.044	0.26	0.887
0.8	1.00	0.144	0.69	0.889	1.00	0.155	0.70	0.901	0.75	0.101	0.26	0.884
0.9	1.00	0.312	0.67	0.895	1.00	0.327	0.69	0.913	0.70	0.210	0.25	0.864
1.0	0.90	0.666	0.64	0.517	1.00	0.691	0.66	0.507	0.00	0.431	0.25	0.540

Table 16: Planted marker, additional quantities (mean over 20 splits): AUC on concordant and flipped test images, gain in flipped AUC from LEACE on the marker, 95th percentile of the random-direction null gain, and AUC of marker presence from the LEACE residual (0.5 means the marker is no longer linearly decodable).

	DINOv2					ResNet-50					xrv				
ρ	conc.	flip	gain	p95	resid.	conc.	flip	gain	p95	resid.	conc.	flip	gain	p95	resid.
0.5	0.868	0.867	-0.000	0.002	0.486	0.884	0.883	-0.002	0.002	0.496	0.896	0.896	-0.002	0.002	0.492
0.6	0.872	0.857	0.013	0.002	0.494	0.891	0.871	0.009	0.002	0.515	0.901	0.887	-0.003	0.004	0.497
0.7	0.885	0.827	0.049	0.003	0.496	0.904	0.840	0.045	0.003	0.510	0.911	0.867	0.011	0.010	0.508
0.8	0.901	0.757	0.123	0.006	0.504	0.923	0.768	0.122	0.008	0.491	0.925	0.824	0.043	0.020	0.498
0.9	0.928	0.616	0.250	0.014	0.492	0.949	0.622	0.258	0.020	0.508	0.943	0.733	0.078	0.045	0.506
1.0	0.967	0.301	0.142	0.035	0.496	0.985	0.295	0.203	0.049	0.514	0.969	0.538	-0.182	0.084	0.515

Appendix J. Eraser Transport, Head Inheritance and CKA

Transport. We fit LEACE in encoder A, map its direction to encoder B through a ridge map, and apply it there, with a random rank-1 eraser as control (Table 17). Transported erasers reduce the linear source probe less than an eraser fit in B itself (DINOv2 to ResNet-50 gives 0.931 against 0.529, and the reverse gives 0.997 against 0.511), the random rank-1 control leaves it at 0.998 to 1.000, and MLP probes decode source at 0.991 or above in every transported condition. The held-out R^2 of the ridge maps is 0.14 to 0.36 (Table 18), so the maps are loose, and the transport result tests a practical transfer and not the existence of a transferable eraser.

Head inheritance. A Belarus/RSNA-trained head from A applied through the map in B scores AUC 0.467 (ResNet-50 from DINOv2), 0.463 (DINOv2 from ResNet-50), 0.499 (xrv from DINOv2) and 0.541 (DINOv2 from xrv) on source-matched data, within the range of the native heads (0.464 to 0.570), and its scores correlate 0.757 to 0.897 with the native B head. The confounded detector is therefore inherited together with the representation, and matched-data TB detection in B is not changed by the own or the transported eraser (Table 18).

CKA. Linear CKA between xrv and DINOv2 is 0.202 and between ResNet-50 and DINOv2 0.408. Erasing the source direction removes 3.2% to 4.5% of the variance and takes the linear source AUC from 1.000 to 0.471 to 0.493, but CKA between an encoder and its erased version stays at 0.890 (ResNet-50), 0.930 (DINOv2) and 0.978 (xrv), and cross-encoder CKA falls by 0.055 and 0.045, against 0.0002 for random rank-1 removals (Table 19). CKA is therefore about 260 to 320 times more sensitive to the source direction than to a random direction, yet in absolute terms it changes little while source decodability collapses. We do not state that CKA is blind to the confound. The result illustrates that a similarity score near 0.9 does not show that a decodable confound is absent (Harvey et al., 2024; Davari et al., 2023), and an audit that relies on representational similarity alone should be supplemented by probes.

Table 17: Eraser transport. A LEACE eraser fit in encoder A is mapped to encoder B through a ridge map. Held-out linear source AUC in B (MLP in parentheses).

A → B	B own LEACE	Transported	Random rank-1
DINOv2 → ResNet-50	0.529 (0.986)	0.931 (0.997)	1.000 (0.999)
ResNet-50 → DINOv2	0.511 (0.991)	0.997 (0.994)	1.000 (0.999)
DINOv2 → xrv	0.524 (0.987)	0.987 (0.991)	0.998 (0.999)
xrv → DINOv2	0.511 (0.991)	0.991 (0.993)	1.000 (0.999)

Table 18: Eraser transport and head inheritance on source-matched data (Shenzhen+Montgomery). A ridge map takes encoder A features to encoder B. Held-out R^2 of the map in each direction, AUC of a Belarus/RSNA-trained head from A applied through the map in B (stitched), AUC of the native heads, correlation of stitched and native B scores, and matched-data TB detection AUC in B with no eraser, B's own LEACE and the transported LEACE.

A → B	R^2 A→B	R^2 B→A	Stitched head	Native head B	Native head A	Corr.	Det. none	Det. own	Det. transp.
DINOv2 → ResNet-50	0.357	0.323	0.467 [0.428, 0.509]	0.464 [0.424, 0.504]	0.471 [0.433, 0.514]	0.78	0.850	0.848	0.844
ResNet-50 → DINOv2	0.323	0.357	0.463 [0.426, 0.503]	0.471 [0.433, 0.514]	0.464 [0.424, 0.504]	0.90	0.839	0.843	0.836
DINOv2 → xrv	0.273	0.140	0.499 [0.461, 0.540]	0.570 [0.531, 0.611]	0.471 [0.433, 0.514]	0.76	0.808	0.815	0.813
xrv → DINOv2	0.140	0.273	0.541 [0.503, 0.582]	0.471 [0.433, 0.514]	0.570 [0.531, 0.611]	0.84	0.839	0.843	0.847

Table 19: Linear centered kernel alignment (CKA) before and after erasing the TB source direction with LEACE (A is the first encoder, B the second). Random rank-1 is the mean over random rank-1 removals.

Pair	Source AUC A, A erased	Source AUC B, B erased	CKA(A,B)	CKA(A, A erased)	CKA(B, B erased)	Variance removed A / B	Decrease in CKA(A,B)	Random rank-1 decrease
ResNet-50 / DINOv2	1.000, 0.471	1.000, 0.493	0.408	0.890	0.930	0.045 / 0.043	0.055	0.0002
xrv / DINOv2	1.000, 0.481	1.000, 0.493	0.202	0.978	0.930	0.032 / 0.043	0.045	0.0002

Appendix K. Supplementary Analyses From the Separate Pipeline

These results come from a separate analysis pipeline and were not recomputed under the protocol of Appendix D. They remain informative and should be read with the caveats given.

- **TB extent.** The sensitivity gap between groups A and B under a pulmonary-only-trained detector was 0.007 (95% CI −0.032 to 0.045). Ablating the extent direction left accuracy unchanged (McNemar $\chi^2 = 0$, $p = 1.0$), and ablating the residual direction lowered accuracy from 0.995 to 0.945 (group A with normals, $n = 2342$) and from 0.994 to 0.943 (group B, $n = 2346$). In the xrv rerun, extent ablation lowered normal-class accuracy from 0.998 to 0.835, so the two pipelines disagree and are not reconciled. Both extent groups are Belarus DICOM, so none of these results can separate extent from source.
- **TB-Intra.** Within the Belarus DICOM cases only ($n = 265$), the extent detector reached AUC 0.726 [0.665, 0.788] and accuracy 0.657. Ablating the classifier boundary gave accuracy 0.506 and a random orthogonal direction left it at 0.657. This bounds complete reliance on format for the extent sub-task and says nothing about the main detection task, which has no matched normal class.
- **INLP trajectory.** A 50-round INLP source-probe AUC fell from 0.951 to 0.928 in this pipeline. The held-out protocol does not reproduce a plateau (xrv 0.882 at 50 rounds, 0.501 at 200; Table 4).
- **CGGA.** The CGGA mRNAseq-693 cohort ($n = 693$, with 507 concordant and 186 discordant patients) gave concordant AUC 1.000 and discordant AUC 0.000, which follows from the cell construction. It is a consistency check and not independent validation.
- **Malaria.** The full-image ResNet-50 classifier had accuracy 0.927 (AUC 0.976). A staining-direction ablation with a saturation-defined direction gives 0.927 vs. 0.928 (Table 5). The RBC-only versus random-crop ordering reverses between the two mask definitions, so no localization statement is made.

Appendix L. Deviations and Analysis Status

- **Pre-registration.** Decision thresholds such as the 0.02 flag margin were set in our analysis plan and were not registered externally.
- **Follow-up selection.** Follow-up tests were chosen after earlier results. No multiplicity correction was applied.

- **Glioma cohort.** The cohort contains the 995 patients with IDH status, which is 1,122 rows minus 127 rows without IDH status.
- **Glioma grade ablation.** Ablating the grade direction yields a constant prediction (Section 5.3, Table 9) and carries no subgroup information.
- **Malaria staining.** The saturation-defined direction leaves accuracy unchanged (0.927 vs. 0.928), so staining is neither supported nor excluded.
- **Malaria localization.** The RBC-only versus random-crop ordering depends on the mask definition.

Appendix M. Limitations

(1) TB positives and RSNA normals differ in both format and source population, and the matched Shenzhen and Montgomery sets differ in population again, so format effects are not isolated. Metadata alone predict TB within the matched sets (AUC 0.748). (2) The xrv encoder is the least sensitive in calibration. (3) The planted marker is a single synthetic corner pattern in TB radiographs, and calibration was not run for glioma or malaria. (4) The glioma domain is tabular, has no age or sex, and the discordant set has 141 patients. (5) Malaria masks are unvalidated, patient groups are parsed from file names, and 24.9% of groups are single-class. (6) Intervals reflect test-sample resampling, with few seeds for ablation controls (2 to 10). (7) The background-only and stain-normalization controls were not run. Grayscale conversion and the cell silhouette view are the closest available checks. (8) Harmonization was applied to xrv features only, and fine-tuned encoders were not studied. (9) Ablation and erasure act on frozen representations and do not identify causes in the clinical data-generating process. (10) We evaluated one flag criterion for the planted-marker audit and applied no multiplicity correction to the many comparisons reported. (11) The TB extent verdict relies mainly on xrv features, for which the calibration shows lower power than for DINOv2 and ResNet-50.

Appendix N. Reproducibility and Provenance

Every table in this appendix and Figures 1 and 3 are generated by scripts from the result files, so no value in them was typed by hand. The main-text numbers are copied from the same files. Result files cover source-matched TB transfer (four feature sets), eraser comparison (four feature sets), ablation controls (TB xrv and DINOv2, malaria ResNet-50 full and RBC views, glioma), glioma refits, patient-grouped malaria views (three encoders and a coarser grouping), planted-marker calibration (three encoders), eraser transport (four encoder pairs) and the malaria patient-state tests (three encoders). The separate-pipeline items listed in Appendices D and K are the only numbers not regenerated.